BIG DATA CHALLENGE: SUSTAINABILITY IN A POST-COVID WORLD

R2 Data Labs

Emergent Alliance

Team M

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Based excluding non-linear scaling, please contact Victoria Dal or Rebecca Sauchman Jones.

The effects of COVID-19 lockdown measures on the supply and demand of electricity in the UK

1 Introduction

A new type of coronavirus and COVID-19, the disease it causes, has been occupying the world's agenda for the last 6 months. As cases spread to the rest of the world, it has put both CEO's in the US and the UK, many countries implemented various forms of measures that the bottom of the healthcare systems, and the UK is a 100 % case. Significantly in the UK, a lockdown was implemented on 23rd March, which has been in place for 2 weeks at the time of writing this report. Although the lockdown measures are different in each country, the common aim is to flatten the curve of infections, to restrictions were lifted in some cases. Many sectors, including the UK, are now starting phased transitions back to normality. Although, the lockdown measures are different in each country, the common aim is to flatten the curve of infections, allowing measures having eased. Total cases so far reported for the UK is 29,474 on 27.03.2020 with 2,352 deaths and the daily case number is 4,450 on 27.03.2020.

In Jung 2019, Ma et al. discuss the first major case in Jung which is occurring in wet and winter of cold so 2019. With the impacts of climate change being the most highly cited long term risk, emerging from Coronavirus has been seen as an opportunity to implement more sustainable measures in the short term, such as a reduction in carbon dioxide and other pollutant emissions, which have been linked to higher mortality rates and exposure to as well as increasing the amount of renewable energy sources. Perhaps there are lessons from the lockdown that can be provide some insight into the short, 10 % but, we will narrow down energy consumption to focus on electricity and investigate how the supply and demand of electricity has changed.

Aims

We have established the following aims that we will achieve:

1. Find out what has been the impact of COVID - 19 on the supply and demand of electricity in the UK.
2. Find out what could have happened to supply and demand of the UK electricity if COVID-19 had not happened.

Objectives

In order to achieve our aims, we will complete the following objectives:

1. Find the UK electricity supply and demand data to examine the effects of COVID-19.
2. Find UK electricity available and demand data to examine the effects of COVID-19 on demand.
3. Find 2020 drilling demand to show high resolution impact of COVID-19 lockdown measures on demand.
4. Find UK electricity demand data to show the impact of COVID-19 lockdown measures on demand.

Structure of this report

Section 2 – Methods explains what methods we have employed and imports the libraries and libraries required.
Section 3 – Impact on electricity demand, we will explore the effects on the electricity demand, accounting for seasonality.
Section 4 – Historical UK electricity supply and demand to the UK during the years and to 2020.
Section 5 – Forecasting UK electricity demand, we will forecast what the demand would be.
Section 6 – Conclusion and longer term impacts on electricity demand in COVID-19 from UK Impact suggestions.

2 Methods

2.1 Libraries and modules

The following packages were needed for the program's original environment (on .app):

- numpy
- pandas
- matplotlib
- sklearn
- xlrd
- xlwt
- openpyxl

The following libraries and modules are required:

```
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
import xlrd
import xlwt
import openpyxl
import datetime as dt
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import cross_val_score
from sklearn.model_selection import RandomizedSearchCV
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from sklearn.model_selection import cross_val_score,cross_val_predict
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
```

2.2 Data collection and selection

2.11 Bridging data

In order to look at the effects that COVID- 19 has had on electricity supply and demand it is necessary to understand the old trends and patterns in the data before the lockdown measures were put in place. We will use the data from the last 2 years to look at trends and changes that habits. To do this we have used the emergency data from the Oxford COVID-19 Government Response Tracker (OxCGRT) (Hale et al. 2020). This data set contains a wide range of indicators to track how the response to COVID-19 has changed over time. We have used the indicators, stringent index that utilizes current and accurate data.

2.2.2 Electricity data

Electricity supply is the amount of electricity that is available while electricity demand is the total amount of electricity that is used at any given moment. Because electricity cannot be stored, it must be produced at the same rate that it is consumed (Ma et al. 2019). Although electricity demand is over-predict is lasting not COVID-19 has affected behaviour and habits. It is important to find out. It's important to find out what the impact COVID-19 has had on electricity demand. Although, we will be using the data to look at the impact COVID-19 has had on the UK's electricity demand, it is important to understand what impact COVID-19 has had on electricity supply and demand. We will be using the data from National Grid ESO (2020).

Electricity demand

Electricity demand is the amount of electricity that is used within 15 of Energy Trends UK (Fraser 2020). This provides data on the amount of electricity which is used across the UK, including the transport, heat and industrial sectors. Although this data set includes products, it also provides data on electricity sold by the major power producers and accounting for the Power and user. Distribution, demand and other. This data set provides a breakdown of the different sectors that electricity is used in and how it has changed.

We have also used a seasonal, quarter and monthly data to examine the data above on a monthly resolution, we investigated alternative sources that could provide higher resolution data on demand during COVID-19 lockdown measures in 2020. We have chosen to use rolling demand data from National Grid ESO (2020). This data set provides a 5 minute resolution on demand across the UK. We have also used the data set that provides a half hourly resolution across the UK, we also chose, we investigated the data in MW/h, to calculate the total rolling demand for the UK and we chose to use it.

2.3 Predictive model

For the predictions we chose the autoregressive integrated moving average (ARIMA) model. This model is fitted to time series data, either to better understand the data or to predict future points in the series.

3. Data impact and checks

3.1 Electricity supply

Total electricity available (by month)

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.1')
df.info()
# Select relevant columns
df = df[['Month', 'Total electricity available (GWh)']]
df.rename(columns={'Total electricity available (GWh)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

Electricity supply by Fuel type

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.2')
df.info()
# Select relevant columns
df = df[['Month', 'Coal', 'Oil', 'Gas', 'Other fuels']]
df.rename(columns={'Coal': 'coal', 'Oil': 'oil', 'Gas': 'gas', 'Other fuels': 'renewables'},)
df.head(3)

# Convert to MW
df['coal'] = df['coal']*1000
df['oil'] = df['oil']*1000
df['gas'] = df['gas']*1000
df['renewables'] = df['renewables']*1000
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

Electricity supply by Fuel type (renewable vs non-renewable)

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.3')
df.info()
# Select relevant columns
df = df[['Month', 'Coal', 'Oil', 'Gas', 'Other fuels']]
df.rename(columns={'Coal': 'coal', 'Oil': 'oil', 'Gas': 'gas', 'Other fuels': 'renewables'},)
df.head(3)

# Convert to MW
df['coal'] = df['coal']*1000
df['oil'] = df['oil']*1000
df['gas'] = df['gas']*1000
df['renewables'] = df['renewables']*1000
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

3.2 Electricity demand

Total electricity sold by month

```
# Import the data
df = pd.read_excel('./data/electricity_demand.xlsx', sheet_name='Q1.1')
df.info()
# Select relevant columns and rows (cut off last three months)
df = df[['Month', 'Total electricity sold (GWh)']][:-3]
df.rename(columns={'Total electricity sold (GWh)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 2 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity sold (GWh)             120 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 1.9 KB
   Month  Total electricity sold  demand
0  2010-01-01                 NaN     NaN
1  2010-02-01                 NaN     NaN
2  2010-03-01                 NaN     NaN
```

Rolling electricity demand – daily average

```
# Import the data
df = pd.read_excel('./data/rolling_demand.xlsx')
# Select relevant columns
df = df[['Settlement date', 'Rolling 24 hour average demand (MW)']]
df.rename(columns={'Rolling 24 hour average demand (MW)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 2 columns):
Settlement date                          1461 non-null  datetime64[ns]
Rolling 24 hour average demand (MW)      1461 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 11.5 KB
   Settlement date  Rolling 24 hour average demand (MW)  demand
0  2020-01-01                              29563.958  29563.958
1  2020-01-02                              29563.958  29563.958
2  2020-01-03                              29563.958  29563.958
```

Rolling electricity demand – weekly average

```
# Import the data
df = pd.read_excel('./data/rolling_demand.xlsx', sheet_name='Q1.2')
df.info()
# Select relevant columns
df = df[['Settlement date', 'Rolling 24 hour average demand (MW)']]
df.rename(columns={'Rolling 24 hour average demand (MW)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 208 entries, 0 to 207
Data columns (total 2 columns):
Settlement date                          208 non-null  datetime64[ns]
Rolling 24 hour average demand (MW)      208 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 3.3 KB
   Settlement date  Rolling 24 hour average demand (MW)  demand
0  2020-01-01                              29563.958  29563.958
1  2020-01-08                              29563.958  29563.958
2  2020-01-15                              29563.958  29563.958
```

3.3 Bridging

```
# Import the data
df = pd.read_excel('./data/oxford_response.xlsx', sheet_name='19_national')
# Select relevant columns
df = df[['Date', 'StringencyIndex']]
df.rename(columns={'StringencyIndex': 'stringency'},)
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 366 entries, 0 to 365
Data columns (total 2 columns):
Date                                     366 non-null  datetime64[ns]
StringencyIndex                          366 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 5.8 KB
   Date  StringencyIndex  stringency
0  2020-01-01             8.33       8.33
1  2020-01-02             8.33       8.33
2  2020-01-03             8.33       8.33
```

4. Historical UK electricity supply and demand prior to COVID-19

4.1 Historical electricity supply

```
# Plot historical data
df.plot(figsize=(12,6))

Text(0.5,0,'Month')
Text(0.5,1,'Total electricity available')
<matplotlib.
<matplotlib.
```

All available data

```
# Plot all data to find out when it's realistic (2010 demand)
df['demand'] = df['demand']*1000
df['demand'] = df['demand']*10

# Set plot style
plt.rcParams['figure.figure=']
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Total monthly electricity available in the UK public supply system')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

The total electricity available detected by supply in the UK public supply system follows a seasonal pattern, peaking in the winter months and decreasing exponentially in the summer months. Since May 2009, the difference between demand and other supply has been increasing, which suggests that the demand for electricity has increased while the electricity supply following a trend over time.

There is a lot of data here so we will have a closer look at our the last 10 years.

Last 10 years

```
# Plot last 10 years
df['demand'] = df['demand']
df['demand'] = df['demand']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Total monthly electricity available in the UK public supply system')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

The UK's data supplied by the pandemic lockdown. We overall trend over the last 10 years has been a declining one and there is a pattern emerging of a decline beginning every year around months.

Last year

```
# Plot last year
df['demand'] = df['demand']
df['demand'] = df['demand']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Total monthly electricity available in the UK public supply system')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```


Looking at electricity supply over 2019 shows an less supply variation across a typical year. The highest supply of electricity is recorded in January and the lowest in August. Although, there is a drop in supply during the Christmas period, which is to be expected. Overall, the amount of supply is at a rate of the lowest, so it is clear that the UK's demand for supply is relatively

4.2 Electricity supply by type of fuel

```
# Plot historical data
df.plot(figsize=(12,6))

Text(0.5,0,'Month')
Text(0.5,1,'Supply')
<matplotlib.
<matplotlib.
```

All available data

```
# Plot all data to find out when it's realistic (2010 demand)
df['coal'] = df['coal']
df['oil'] = df['oil']
df['gas'] = df['gas']
df['renewables'] = df['renewables']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Supply')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

Total monthly electricity available in the UK public supply system by fuel type in 2010

In the late 1990s and early 2000s, these types of fuel were used to generate the majority of the UK electricity supply. Coal, gas, renewables and oil have been used to generate the majority of the UK electricity supply. Coal has been used the most and has been used to generate the majority of the UK's electricity.

Last 10 years

```
# Plot last 10 years
df['coal'] = df['coal']
df['oil'] = df['oil']
df['gas'] = df['gas']
df['renewables'] = df['renewables']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Supply')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

Total monthly electricity available in the UK public supply system by fuel type in 2019

Looking at electricity supply over 2019 shows an less supply variation across a typical year. The highest supply of electricity is recorded in January and the lowest in August. Although, there is a drop in supply during the Christmas period, which is to be expected. Overall, the amount of supply is at a rate of the lowest, so it is clear that the UK's demand for supply is relatively

4.3 Electricity supply by fuel type (renewable vs non-renewable)

```
# Plot historical data
df.plot(figsize=(12,6))

Text(0.5,0,'Month')
Text(0.5,1,'Supply')
<matplotlib.
<matplotlib.
```

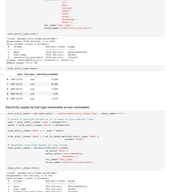
All available data

```
# Plot all data to find out when it's realistic (2010 demand)
df['coal'] = df['coal']
df['oil'] = df['oil']
df['gas'] = df['gas']
df['renewables'] = df['renewables']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Supply')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

Total monthly electricity available in the UK public supply system by fuel type in 2010

Last 10 years

```
# Plot last 10 years
df['coal'] = df['coal']
df['oil'] = df['oil']
df['gas'] = df['gas']
df['renewables'] = df['renewables']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Supply')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```



Total monthly electricity available in the UK public supply system by fuel type in 2019

Last year

```
# Plot last year
df['coal'] = df['coal']
df['oil'] = df['oil']
df['gas'] = df['gas']
df['renewables'] = df['renewables']

# Set plot
ax = df.plot(figsize=(12,6))
ax.set_ylabel('Demand (MW)')
ax.set_xlabel('Month')
ax.set_title('Supply')
plt.rcParams['axes.grid=']
plt.rcParams['axes.grid.axis=']
plt.rcParams['axes.grid.which=']
plt.rcParams['axes.grid.linewidth=']
plt.rcParams['axes.grid.linestyle=']
plt.rcParams['axes.grid.color=']
plt.rcParams['axes.grid.alpha=']
plt.rcParams['axes.grid.minor.linewidth=']
plt.rcParams['axes.grid.minor.linestyle=']
plt.rcParams['axes.grid.minor.color=']
plt.rcParams['axes.grid.minor.alpha=']
```

Total monthly electricity available in the UK public supply system by fuel type in 2019

5 Forecasting UK electricity supply and demand prior to COVID-19

5.1 Historical electricity supply

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.1')
df.info()
# Select relevant columns
df = df[['Month', 'Total electricity available (GWh)']]
df.rename(columns={'Total electricity available (GWh)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

Electricity supply by Fuel type

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.2')
df.info()
# Select relevant columns
df = df[['Month', 'Coal', 'Oil', 'Gas', 'Other fuels']]
df.rename(columns={'Coal': 'coal', 'Oil': 'oil', 'Gas': 'gas', 'Other fuels': 'renewables'},)
df.head(3)

# Convert to MW
df['coal'] = df['coal']*1000
df['oil'] = df['oil']*1000
df['gas'] = df['gas']*1000
df['renewables'] = df['renewables']*1000
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

Electricity supply by Fuel type (renewable vs non-renewable)

```
# Import the data
df = pd.read_excel('./data/electricity_supply.xlsx', sheet_name='Q1.3')
df.info()
# Select relevant columns
df = df[['Month', 'Coal', 'Oil', 'Gas', 'Other fuels']]
df.rename(columns={'Coal': 'coal', 'Oil': 'oil', 'Gas': 'gas', 'Other fuels': 'renewables'},)
df.head(3)

# Convert to MW
df['coal'] = df['coal']*1000
df['oil'] = df['oil']*1000
df['gas'] = df['gas']*1000
df['renewables'] = df['renewables']*1000
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity available (GWh)        120 non-null  float64
Total demand (GWh)                       120 non-null  float64
Net imports (GWh)                        120 non-null  float64
Coal                                     120 non-null  float64
Oil                                      120 non-null  float64
Gas                                      120 non-null  float64
Other fuels                              120 non-null  float64
dtypes: float64(7), datetime64[ns](1)
memory usage: 7.5 KB
   Month  Total electricity available  demand
0  2010-01-01                 29.000  29000.0
1  2010-02-01                 27.000  27000.0
2  2010-03-01                 29.000  29000.0
```

5.2 Electricity demand

Total electricity sold by month

```
# Import the data
df = pd.read_excel('./data/electricity_demand.xlsx', sheet_name='Q1.1')
df.info()
# Select relevant columns and rows (cut off last three months)
df = df[['Month', 'Total electricity sold (GWh)']][:-3]
df.rename(columns={'Total electricity sold (GWh)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 2 columns):
Month                                    120 non-null  datetime64[ns]
Total electricity sold (GWh)             120 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 1.9 KB
   Month  Total electricity sold  demand
0  2010-01-01                 NaN     NaN
1  2010-02-01                 NaN     NaN
2  2010-03-01                 NaN     NaN
```

Rolling electricity demand – daily average

```
# Import the data
df = pd.read_excel('./data/rolling_demand.xlsx')
# Select relevant columns
df = df[['Settlement date', 'Rolling 24 hour average demand (MW)']]
df.rename(columns={'Rolling 24 hour average demand (MW)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 2 columns):
Settlement date                          1461 non-null  datetime64[ns]
Rolling 24 hour average demand (MW)      1461 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 11.5 KB
   Settlement date  Rolling 24 hour average demand (MW)  demand
0  2020-01-01                              29563.958  29563.958
1  2020-01-02                              29563.958  29563.958
2  2020-01-03                              29563.958  29563.958
```

Rolling electricity demand – weekly average

```
# Import the data
df = pd.read_excel('./data/rolling_demand.xlsx', sheet_name='Q1.2')
df.info()
# Select relevant columns
df = df[['Settlement date', 'Rolling 24 hour average demand (MW)']]
df.rename(columns={'Rolling 24 hour average demand (MW)': 'demand'},)
df.head(3)

# Fix dtypes
df['demand'] = df['demand']*1000

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 208 entries, 0 to 207
Data columns (total 2 columns):
Settlement date                          208 non-null  datetime64[ns]
Rolling 24 hour average demand (MW)      208 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 3.3 KB
   Settlement date  Rolling 24 hour average demand (MW)  demand
0  2020-01-01                              29563.958  29563.958
1  2020-01-08                              29563.958  29563.958
2  2020-01-15                              29563.958  29563.958
```

5.3 Bridging

```
# Import the data
df = pd.read_excel('./data/oxford_response.xlsx', sheet_name='19_national')
# Select relevant columns
df = df[['Date', 'StringencyIndex']]
df.rename(columns={'StringencyIndex': 'stringency'},)
df.head(3)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 366 entries, 0 to 365
Data columns (total 2 columns):
Date                                     366 non-null  datetime64[ns]
StringencyIndex                          366 non-null  float64
dtypes: float64(1), datetime64[ns](1)
memory usage: 5.8 KB
   Date  StringencyIndex  stringency
0  2020-01-01             8.33       8.33
1  2020-01-02             8.33       8.33
2  2020-01-03             8.33       8.33
```

6 Conclusion and longer term impacts on electricity demand in COVID-19 from UK Impact suggestions

In the 2000s, Ma et al. discuss the first major case in Jung which is occurring in wet and winter of cold so 2019. With the impacts of climate change being the most highly cited long term risk, emerging from Coronavirus has been seen as an opportunity to implement more sustainable measures in the short term, such as a reduction in carbon dioxide and other pollutant emissions, which have been linked to higher mortality rates and exposure to as well as increasing the amount of renewable energy sources. Perhaps there are lessons from the lockdown that can be provide some insight into the short, 10 % but, we will narrow down energy consumption to focus on electricity and investigate how the supply and demand of electricity has changed.

Aims

We have established the following aims that we will achieve:

1. Find out what has been the impact of COVID - 19 on the supply and demand of electricity in the UK.
2. Find out what could have happened to supply and demand of the UK electricity if COVID-19 had not happened.

Objectives

In order to achieve our aims, we will complete the following objectives:

1. Find the UK electricity supply and demand data to examine the effects of COVID-19.
2. Find UK electricity available and demand data to examine the effects of COVID-19 on demand.
3. Find 2020 drilling demand to show high resolution impact of COVID-19 lockdown measures on demand.
4. Find UK electricity demand data to show the impact of COVID-19 lockdown measures on demand.

Structure of this report

Section 2 – Methods explains what methods we have employed and imports the libraries and libraries required.
Section 3 – Impact on electricity demand, we will explore the effects on the electricity demand, accounting for seasonality.
Section 4 – Historical UK electricity supply and demand to the UK during the years and to 2020.
Section 5 – Forecasting UK electricity demand, we will forecast what the demand would be.
Section 6 – Conclusion and longer term impacts on electricity demand in COVID-19 from UK Impact suggestions.

2 Methods

2.1 Libraries and modules

The following packages were needed for the program's original environment (on .app):

- numpy
- pandas
- matplotlib
- sklearn
- xlrd
- xlwt
- openpyxl

The following libraries and modules are required:

```
import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
import xlrd
import xlwt
import openpyxl
import datetime as dt
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import cross_val_score
from sklearn.model_selection import RandomizedSearchCV
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from sklearn.model_selection import cross_val_score,cross_val_predict
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_score
```

2.2 Data collection and selection

2.11 Bridging data

In order to look at the effects that COVID- 19 has had on electricity supply and demand it is necessary to understand the old trends and patterns in the data before the lockdown measures were put in place. We will use the data from the last 2 years to look at trends and changes that habits. To do this we have used the emergency data from the Oxford COVID-19 Government Response Tracker (OxCGRT) (Hale et al. 2020). This data set contains a wide range of indicators to track how the response to COVID-19 has changed over time. We have used the indicators, stringent index that utilizes current and accurate data.

2.2.2 Electricity data

Electricity supply is the amount of electricity that is available while electricity demand is the total amount of electricity that is used at any given moment. Because electricity cannot be stored, it must be produced at the same rate that it is consumed (Ma et al. 2019). Although electricity demand is over-predict is lasting not COVID-19 has affected behaviour and habits. It is important to find out. It's important to find out what the impact COVID-19 has had on electricity demand. Although, we will be using the data to look at the impact COVID-19 has had on the UK's electricity demand, it is important to understand what impact COVID-19 has had on electricity supply and demand. We will be using the data from National Grid ESO (2020).

Electricity demand

Electricity demand is the amount of electricity that is used within 15 of Energy Trends UK (Fraser 2020). This provides data on the amount of electricity which is used across the UK, including the transport, heat and industrial sectors. Although this data set includes products, it also provides data on electricity sold by the major power producers and accounting for the Power and user. Distribution, demand and other. This data set provides a breakdown of the different sectors that electricity is used in and how it has changed.

We have also used a seasonal, quarter and monthly data to examine the data above on a monthly resolution, we investigated alternative sources that could provide higher resolution data on demand during COVID-19 lockdown measures in 2020. We have chosen to use rolling demand data from National Grid ESO (2020). This data set provides a 5 minute resolution on demand across the UK. We have also used the data set that provides a half hourly resolution across the UK, we also chose, we investigated the data in MW/h, to calculate the total rolling demand for the UK and we chose to use it.

2.3 Predictive model

For the predictions we chose the autoregressive integrated moving average (ARIMA) model. This model is fitted to time series data, either to better understand the data or to predict future points in the series.

3. Data impact and checks

3.1 Electricity supply

Total electricity available (by month)

```
# Import the data
df = pd.read_excel
df.info()
# Select relevant columns
df = df[
df.rename(
df.head(3)

# Fix dtypes
df[
df[

<class
RangeIndex:
Data columns (total 8 columns):
Month                                    120 non-null  
```

Looking at the changes over a year, electricity provided by different types of fuel is relatively stable. There is a decrease in the supply of electricity from coal and an increase in the supply of electricity from other renewable components and around the time and after the end of energy crisis, the summer month.

4.3 Electricity supply by type of fuel (renewable vs non-renewable)

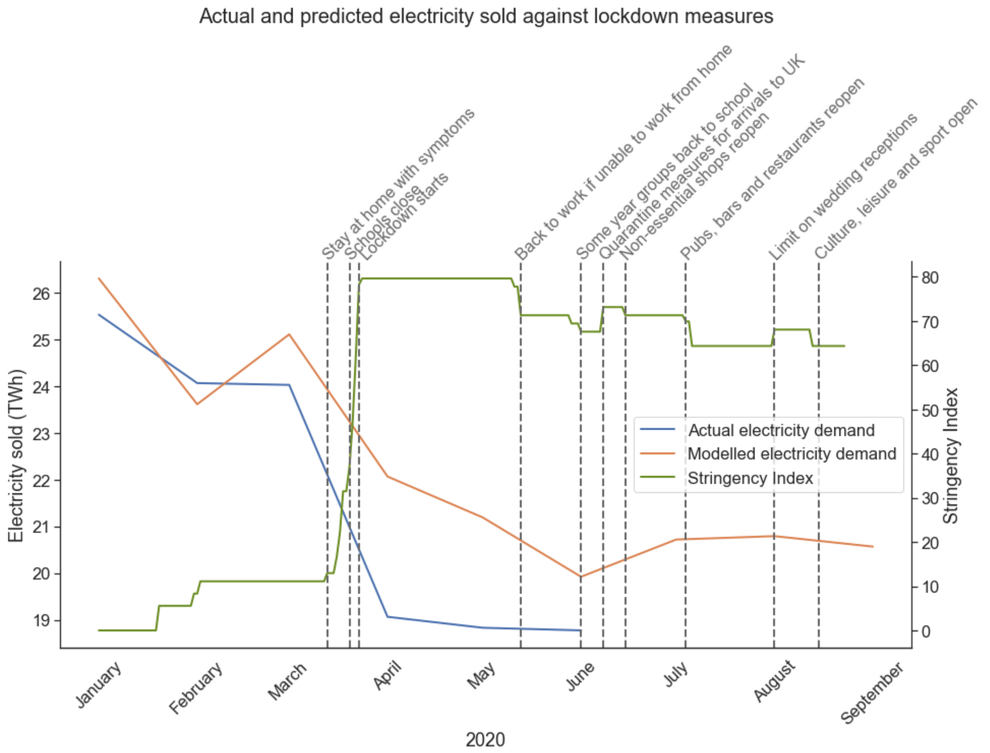
```
# Monthly data by type of fuel and year
monthly_data = df.groupby(['year', 'month', 'type'])
[0]
[0]
[0]
```

```
# head(df_ts)
# tail(df_ts)
# sample_frac(df_ts)
# glimpse(df_ts)
# skim(df_ts)
```

```
# df_ts$Date = as.Date(df_ts$Date)
# ggplot(df_ts, aes(x = Date, y = electricity_demand)) +
# geom_line(size = 0.2, alpha = 0.5)
```

Steady average-long electricity demand during 2020 COVID-19 lockdowns
electricity_demand
Emergency holiday
Stay at home
Shop locally
Work remotely
Work from home
Lockdown
Lockdown
Lockdown

```
leg = ax2.legend(lines + lines2, labels + labels2,
                 frameon=True,
                 loc='center right')

# Define the date format.
date_form = DateFormatter('%B')
ax2.xaxis.set_major_formatter(date_form)

# Plot annotations.
title = 'Actual and predicted electricity sold against lockdown measures'

plot_dates(84, ax2, title);
```



This final plot shows that the timing of the decrease in electricity demand in 2020 does seem to coinside with what would have been expected to happen in 2020, a decline beginning in March and continuing to June. However, the decline occured much more quickly than the model predicted and it falls to lower levels than the model predicted. We can infer from this that although we would have expected electricity demand to fall around this time, due to the COVID-19 lockdown measures, electricity demand fell more than would have been expected had COVID-19 not happened. It would be interesting to add the data from the rest of the year once it is available to see if demand increases from June as the model has predicted. A next step while we are waiting for that data to be published would be to plot the rolling demand against the model and see if that correlates.

---

7. Discussion

People's behavior and electricity

The analysis of electricity demand trend changes due to the coronavirus pandemic can assist companies in formulating major changes in cultures, habits, lifestyle options and choices in the short-term and long-term future. The fluctuations that are observed in the demand are a result of the changes in people's behavior and lifestyle due to the government's coronavirus actions. Wilson et al. (2020) state that some major changes that have had an impact in electricity demand are the following:

- Restrictions in international travel and limitations in transportation lower energy demand as well as declining carbon emissions.
- Remote working has caused weekdays to look like weekends when it comes to the electricity demand. Since many people work or study from home, the typical morning electricity "peak" has flattened out. On the other hand, the digital traffic has increased a lot, which is expected to increase the load on the electrical system.

Both the government and corporations can use this information to prepare for a potential new "normal" , where working and studying remotely will be a cultural and habitual change. This will have an influence on the energy consumption throughout the week by transferring demand peaks to earlier parts of the week.

Future work

We could continue our research by:

- Trying out different prediction models to increase accuracy.
- Analysing the economics data to figure out how everything is correlated and the effect energy supply and demand has affected different industries.
- Work on a model that will be able to combine all this data and come up with conclusions that can be valuable for both the government and corporations and hopefully assist in a more sustainable future.

---

8. Conclusions

- Electricity demand decreased as lockdown measures were put in place.
- The historical data showed a seasonal variation in the demand of electricity with a decline idemand over the summer months, likely due to warmer weather and longer daylight hours.
- The historical data also showed a declining trend in electricity demand since around 2008.
- The model shows that the decrease in demand this year was greater than would have been expected had these same trends continued into 2020.

References

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